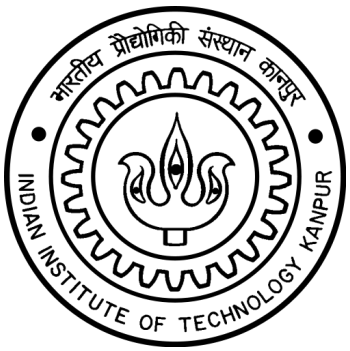


SPA617

RADIO ASTRONOMY LABORATORY

VSWR and finding the Antenna Operating Frequency

RF Transmission, Reflections, Standing Waves

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1. Introduction

1.1. My understanding of VSWR

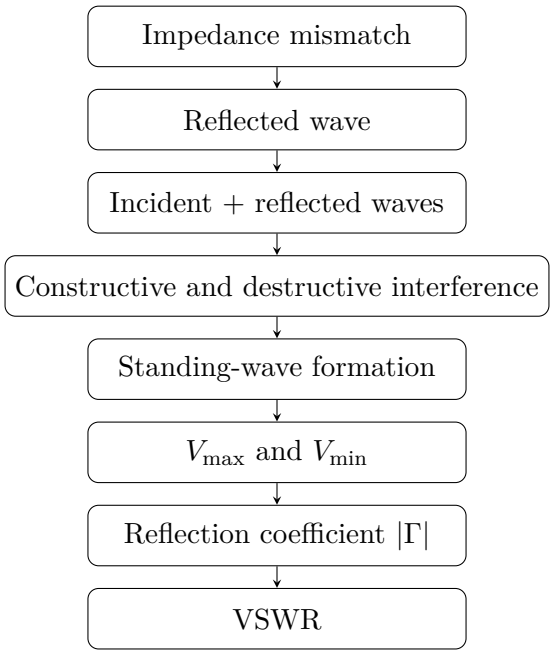
Previously, I was unfamiliar with the concept of VSWR. However, performing the experiments helped me understand how impedance mismatches at different junctions in an RF circuit result in wave reflections and the subsequent formation of standing waves. VSWR provides a quantitative measure of the extent of these standing-wave variations through the ratio of the maximum and minimum voltages along the transmission line.

$$\mathrm{VSWR} = \frac{V_{\max}}{V_{\min}} = \frac{1 + |\Gamma|}{1 - |\Gamma|}, \tag{1}$$

where

$$|\Gamma| = \frac{V_{\max} - V_{\min}}{V_{\max} + V_{\min}}. \tag{2}$$

Understanding VSWR is therefore important in studying RF systems, particularly for long-wavelength detection, where impedance mismatches can lead to reflected power and reduced signal transfer.



1.2. Transmission Lines and Coaxial Cables

Transmission lines are structures that guide electromagnetic waves from one point to another while controlling their propagation and energy transfer. A normal wire cannot always be treated as an ideal connection at high frequencies, since the electromagnetic disturbance takes a finite time to propagate along its length. For example, at a frequency of (100,MHz), the wavelength of an electromagnetic wave in free space is

$$\lambda = \frac{c}{f} = \frac{3 \times 10^8}{10^8} = 3, \mathrm{m}. \tag{3}$$

Therefore, even a wire of length (1m) is a significant fraction of the wavelength and can no longer be treated simply as a lumped circuit element. A transmission line provides a structure that guides the electromagnetic energy along its length.

In this experiment, a coaxial cable was used as the transmission line. It consists of two concentric conductors separated by a dielectric material, with the conductors maintained at different potentials. This geometry gives rise to a distributed capacitance between the conductors along the length of the cable. Similarly, the currents flowing through the conductors produce magnetic fields and hence distributed inductive effects. These distributed capacitance and inductance determine important properties of the transmission line, such as its characteristic impedance and propagation velocity.

The electromagnetic wave propagating through the cable arises from the coupled electric and magnetic fields associated with the varying voltage and current. Consequently, the propagation velocity in the cable is generally different from the speed of light in vacuum.

1.3. Understanding from Viva

1.3.1 Question: What is the necessity of using a quarter-wavelength antenna?

A monopole antenna has two ends: a free (open) end and another end connected to the feed point. Physically, we know that at the open end, the net current must be equal to zero. Therefore, the reflected current wave must be out of phase by π with the incident current wave at the open end. This reflection further leads to the formation of standing waves along the antenna.

The role of the quarter-wavelength is to ensure that the current at the feed point is maximum due to constructive superposition of the incident and reflected waves. To understand this, consider an antenna of length L. The phase change associated with propagation over a distance L is

$$\Delta\phi = kL, \tag{4}$$

where

$$k = \frac{2\pi}{\lambda}. \tag{5}$$

For a quarter-wavelength antenna,

$$L = \frac{\lambda}{4}, \tag{6}$$

and hence the phase accumulated during propagation over the antenna length is

$$\Delta\phi = kL = \frac{2\pi}{\lambda}\frac{\lambda}{4} = \frac{\pi}{2} = 90^\circ. \tag{7}$$

However, the reflected wave travels from the feed point to the open end and then back to the feed point. Therefore, the total propagation phase difference is

$$2kL = 2\left(\frac{\pi}{2}\right) = \pi. \tag{8}$$

Combining this 180° phase change due to propagation with the 180° phase reversal of the current upon reflection at the open end gives

$$\Delta\phi_{\text{total}} = \pi + \pi = 2\pi. \tag{9}$$

Thus, at the feed point, the incident and reflected current waves are in phase and undergo constructive superposition, resulting in maximum current. Hence, a quarter-wavelength antenna provides the required condition for maximum current at the feed point while maintaining zero current at the open end.

1.3.2 Question: For an antenna with fixed length, how would you evaluate what the resonance frequency is?

Experimentally, the antenna length is kept fixed and the frequency is varied while monitoring the feed-point response. The resonance frequency is identified as the frequency at which the feed current/voltage is maximum.

$$f_r = \text{frequency at which } I_{\text{feed}} \text{ is maximum} \tag{10}$$

2. (cont.) Experiment 1 : Antenna characterization

The measured results for the theoretical resonance frequency of 102.74 MHz and an arbitrary comparison frequency of 160 MHz are shown in Table 1. At 102.74 MHz, the voltage varied from 0.7 mV to 1.75 V within the selected ±10 MHz frequency range. This results in a very large VSWR of approximately 2500 and a return loss of only 0.00695 dB.

In comparison, at 160 MHz, the voltage varied between 0.36 V and 1.07 V, giving a much lower VSWR of approximately 2.97 and a return loss of 6.09 dB. The large VSWR obtained around the theoretical resonance indicates that the measured voltage changes very strongly within the frequency sweep around this frequency. Away from the resonance, at 160 MHz, the voltage variation is considerably smaller, resulting in a lower measured VSWR.

Table 1: Measured VSWR and return loss at selected frequencies.

Frequency (MHz)	$V_{\max}$ (V)	$V_{\min}$ (V)	VSWR	Return Loss (dB)
102.74	1.75	0.0007	2500.00	0.00695
160.00	1.07	0.36	2.97	6.09

It should be noted that the $V_{\max}$ and $V_{\min}$ used here are the maximum and minimum voltage amplitudes obtained during the frequency sweep around each selected test frequency, as specified in the experimental procedure.

Note that the following formulae were used for the evalaution of the given parameter in table 1

$$\text{VSWR} = \frac{V_{\max}}{V_{\min}} \tag{11}$$

$$|\Gamma| = \frac{\text{VSWR} - 1}{\text{VSWR} + 1} = \frac{V_{\max} - V_{\min}}{V_{\max} + V_{\min}} \tag{12}$$

$$RL = -20 \log_{10} |\Gamma| \tag{13}$$

3. Aim of Experiment 2 : RF Transmission, Reflections, Standing Waves

To understand signal propagation, reflection, and standing-wave formation in a coaxial transmission line at radio frequencies, and to determine important cable parameters experimentally.

4. Objectives of the Experiment

1. To determine the phase velocity and velocity factor of a signal propagating through a coaxial cable from the frequency spacing between consecutive maxima or minima produced by reflection from an open-ended cable.

2. To study signal reflection from an open-ended cable and from a cable terminated with a $50\,\Omega$ load, and to determine the corresponding reflection coefficients.
3. To determine the inductance and capacitance per unit length of the coaxial cable using the measured phase velocity and the approximate characteristic impedance $Z_0 \approx 50\,\Omega$.
4. To measure the insertion loss and attenuation per unit length of the coaxial cable.

5. Apparatus

The following appartus was used for the experiment:

- RF signal generator
- Digital storage oscilloscope (DSO)
- Spectrum Analyzer
- RF Tee connector
- Coaxial cable under test
- Short reference coaxial cable
- $50\,\Omega$ termination

6. Theory

6.1. Transmission Lines

A transmission line is a structure used to guide electromagnetic energy from one point to another. At radio frequencies, the wavelength of the electromagnetic signal can become comparable to the physical dimensions of the circuit. In such a situation, the propagation of the signal along the conductor cannot be neglected and the circuit can no longer be treated simply as an ideal wire where the ends of the wire can be though of as being maintained at constant potentials.

For example, for a signal of frequency 100 MHz, the wavelength in free space is

$$\lambda = \frac{c}{f} = \frac{3 \times 10^8}{10^8} = 3\,\text{m}. \tag{14}$$

Therefore, a cable of length 2 m represents a significant fraction of the wavelength. The phase change along the cable is given by

$$\Delta\phi = kL = \frac{2\pi L}{\lambda}. \tag{15}$$

For $L = 2\,\text{m}$ and $\lambda = 3\,\text{m}$,

$$\Delta\phi = \frac{2\pi}{3}(2) = \frac{4\pi}{3} = 240^\circ. \tag{16}$$

Thus, the voltage and current cannot be considered to have the same phase throughout the cable. The cable must instead be treated as a transmission line having distributed electrical properties.

In this experiment, a **coaxial cable** is used as the transmission line. It consists of two conductors separated by a dielectric material. The geometry of the cable gives rise to distributed capacitance and inductance along its length. The coaxial cable supports transverse electromagnetic

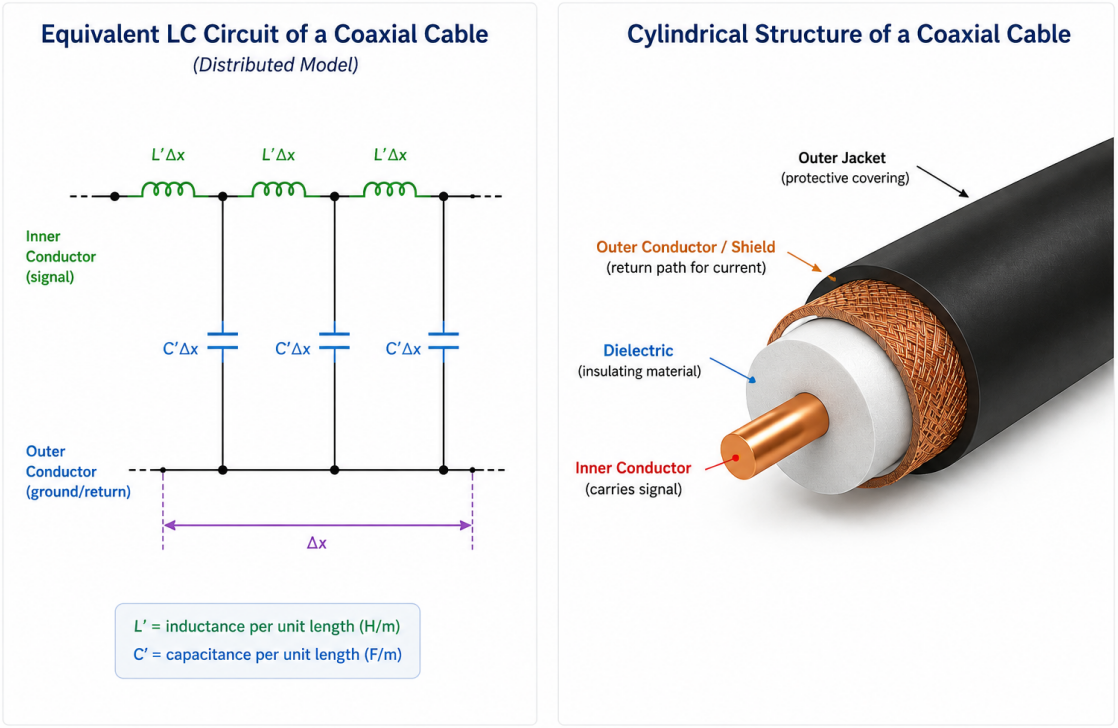


Figure 1: Physical structure and distributed LC model of a coaxial cable.

(TEM) propagation, in which the electric and magnetic fields are transverse to the direction of propagation. The presence of the dielectric also reduces the propagation velocity below the speed of light in vacuum.

6.2. Coaxial Cable and Distributed Parameters

A coaxial cable consists of two concentric conductors separated by a dielectric material. The inner conductor carries the signal, while the outer conductor acts as the return path for the reflected signal and also provides shielding. The dielectric separates the conductors and confines the electromagnetic field predominantly to the region between them. The coaxial geometry allows the cable to support transverse electromagnetic (TEM) propagation.

The potential difference between the inner and outer conductors gives rise to a distributed capacitance per unit length, denoted by C', while the current flowing through the conductors produces a magnetic field and hence a distributed inductance per unit length, denoted by L'. Thus, the coaxial cable can be represented by a distributed LC network.

For a low-loss transmission line, the phase velocity and characteristic impedance are related to these distributed parameters by

$$v_p = \frac{1}{\sqrt{L'C'}} \tag{17}$$

and

$$Z_0 = \sqrt{\frac{L'}{C'}}. \tag{18}$$

6.3. Standing Wave Formation

When a coaxial cable is connected to a load, the impedance at the termination may differ from the characteristic impedance Z_0 of the cable. This impedance mismatch causes a part of the incident wave to be reflected back towards the source. The reflection coefficient is given by

$$\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0}, \tag{19}$$

where Z_L is the load impedance. The incident and reflected waves then superpose, producing constructive and destructive interference and hence a standing-wave pattern.

For certain relative phases, the two waves interfere constructively, while for a phase difference of 180°, they interfere destructively. As the frequency is varied, the phase accumulated during propagation changes, causing the measured amplitude to vary periodically.

Ideally, when the load impedance matches the characteristic impedance of the cable,

$$Z_L = Z_0, \tag{20}$$

the reflection coefficient becomes zero and no standing-wave variation due to reflection is produced.

6.4. Phase Velocity and Velocity Factor

When a signal propagates through a coaxial cable of length ℓ, the reflected wave travels a round-trip distance of 2ℓ. The corresponding round-trip time delay is

$$\tau = \frac{2\ell}{v_p}, \tag{21}$$

where v_p is the phase velocity of the signal in the cable. The round-trip phase accumulated by the reflected wave is therefore

$$\phi_{\mathrm{RT}} = \frac{4\pi f \ell}{v_p} + \phi_\Gamma, \tag{22}$$

where ϕ_Γ is the phase shift introduced upon reflection at the termination. For an ideal open circuit, $\phi_\Gamma = 0$.

As the frequency is varied, the round-trip phase changes. Two consecutive maxima or two consecutive minima in the measured amplitude correspond to a change in round-trip phase of 2π. Hence,

$$\Delta f \, \tau = 1, \tag{23}$$

$$v_p = 2\ell \Delta f. \tag{24}$$

The velocity factor of the cable is defined as the ratio of the phase velocity in the cable to the speed of light in vacuum,

$$\mathrm{VF} = \frac{v_p}{c}, \tag{25}$$

where $c \approx 3 \times 10^8\,\mathrm{m/s}$.

6.5. Reflection Coefficient and Characteristic Impedance

The reflection coefficient quantifies the fraction of the incident wave that is reflected due to an impedance mismatch between the load impedance Z_L and the characteristic impedance Z_0 of the transmission line. It is given by

$$\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0}. \tag{26}$$

When $Z_L = Z_0$, $\Gamma = 0$ and there is no reflection, indicating an ideally matched transmission line. Ideally, to minimize losses we want to set the reflection coefficient to effectively nearly zero.

6.6. Inductance and Capacitance per Unit Length

The distributed inductance L' and capacitance C' describe the inductive and capacitive properties of the coaxial cable per unit length. For a low-loss transmission line, they are related to the phase velocity and characteristic impedance by

$$L' = \frac{Z_0}{v_p}, \qquad C' = \frac{1}{Z_0 v_p}. \tag{27}$$

These formulae come directly from the expressions of impedance and phase velocity in the form of inductance and capacitance per unit length.

6.7. Attenuation

Cable attenuation represents the reduction in signal level as the electromagnetic wave propagates through the cable. The insertion loss is obtained by comparing the measured reference power with the power measured after inserting the cable,

$$IL = P_{\text{ref}} - P_{\text{cable}}, \tag{28}$$

and the attenuation per unit length is given by

$$\alpha = \frac{IL}{\ell}, \tag{29}$$

in dB/m.

Cable attenuation occurs because a portion of the electromagnetic energy is dissipated during propagation, primarily due to the finite conductivity of the conductors and dielectric losses. Consequently, the signal power decreases with the length of the cable.

6.8. Experimental Setup

The experimental setup consists of a 92 cm long coaxial cable, which is used as the primary device under test. The cable is connected to an RF Tee connector, through which the incident signal from the signal generator is fed into the cable. The oscilloscope is connected to the Tee junction to measure the voltage amplitude at this fixed point. For the open-ended cable, the incident wave is reflected at the far end and travels back towards the Tee junction. By varying the frequency of the incident signal, the round-trip phase of the reflected wave changes, resulting in periodic variations in the measured amplitude at the Tee junction. These variations are used to study the reflection and standing-wave behaviour of the coaxial cable. The same experimental setup is then used to compare and evaluate the characteristic impedance of the wire by connecting a load of 50Ω across the coaxial cable.

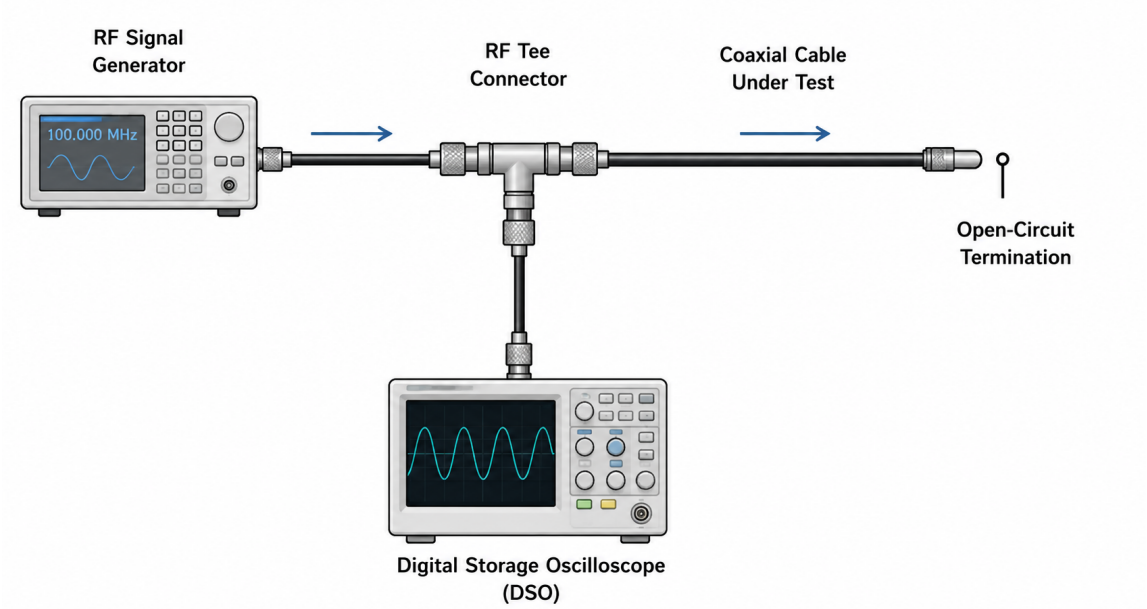


Figure 2: Experimental setup for studying signal propagation and reflection in the coaxial cable.

7. Observations

The physical length of the coaxial cable was measured as

$$\ell = 0.92\,\text{m}. \tag{30}$$

7.1. Part A: Phase Velocity and Velocity Factor

The coaxial cable was left open at the far end, and the frequency of the signal generator was varied while recording the corresponding voltage amplitude at the Tee junction. The measured data are tabulated in table 3.

7.2. Part B: Connecting the 50Ω Load

The values for V_{max} and V_{min} when using the load are shown in table 2

Table 2: Measured voltage amplitudes for different cable terminations.

Termination	V_{max} (V)	V_{min} (V)
50 Ω load	1.09	0.881

Table 3: Measured voltage amplitude as a function of frequency for the open-ended coaxial cable.

Frequency (MHz)	Voltage Amplitude (V)
57	0.060
62	0.225
68	0.578
70	0.693
75	0.974
80	1.28
85	1.51
90	1.61
95	1.71
97	1.72
100	1.69
105	1.65
110	1.65
115	1.58
120	1.54
125	1.50
130	1.42
135	1.33
140	1.23
145	1.08
150	0.932
155	0.762
160	0.579
165	0.395
170	0.200
171	0.163
172	0.121
173	0.090
174	0.080
175	0.097

7.3. Part D: Attenuation and Insertion Loss

The attenuation and insertion loss were evaluated at multiple frequencies for figuring out at which frequency are the losses and attenuation per unit length minimum. The results, along with the data are given in table 4

Table 4: Measured insertion loss and attenuation of the coaxial cable.

Frequency (GHz)	P_{ref} (dBm)	P_{cable} (dBm)	IL (dB)	α (dB/m)
1.0	20.00	17.45	2.55	2.77
1.6	20.00	17.00	3.00	3.26
0.5	20.00	18.50	1.50	1.63
2.0	20.00	16.49	3.51	3.82

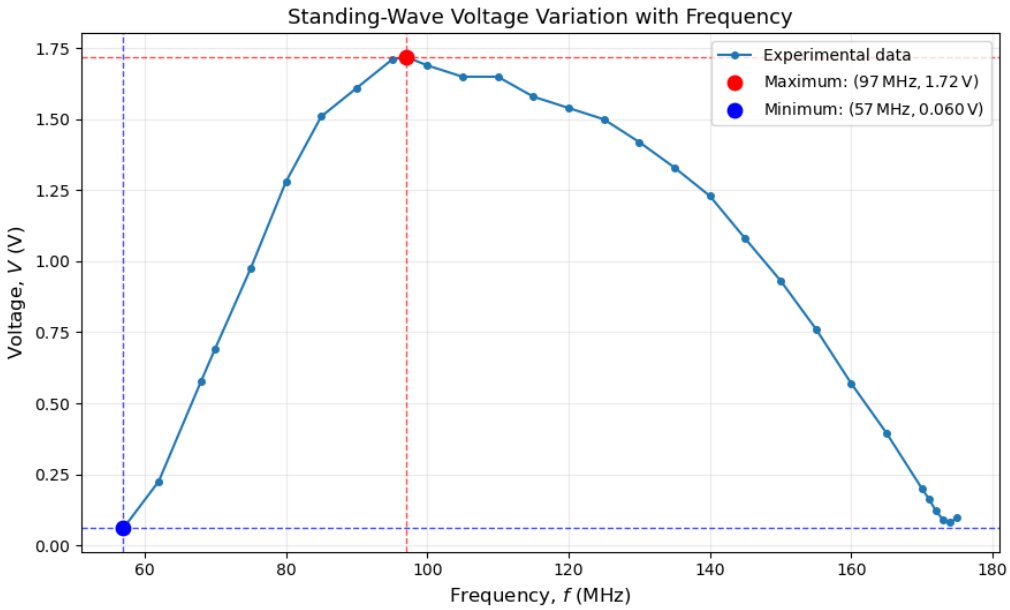


Figure 3: Measured voltage amplitude as a function of frequency for the open-ended coaxial cable.

7.4. Calculations

7.4.1 Phase Velocity

The length of the coaxial cable used in the experiment was

$$\ell = 92\,\text{cm} = 0.92\,\text{m}. \tag{31}$$

From the frequency-amplitude measurements, two consecutive minima were observed at

$$f_1 = 57\,\text{MHz} \tag{32}$$

$$f_2 = 174\,\text{MHz}. \tag{33}$$

Therefore, the frequency separation between consecutive minima is

$$\Delta f = f_2 - f_1 = 174 - 57 = 117\,\text{MHz}. \tag{34}$$

For an open-ended cable, the phase velocity is given by

$$v_p = 2\ell\Delta f. \tag{35}$$

$$v_p = 2(0.92)(117 \times 10^6) \tag{36}$$

$$= 2.1528 \times 10^8\,\text{m/s}. \tag{37}$$

$$\boxed{v_p = 2.153 \times 10^8\,\text{m/s}}. \tag{38}$$

7.4.2 Velocity Factor

The velocity factor of the cable is defined as

$$\mathrm{VF} = \frac{v_p}{c}, \tag{39}$$

where $c = 3 \times 10^8\,\mathrm{m/s}$ is the speed of light in vacuum.
Therefore,

$$\mathrm{VF} = \frac{2.1528 \times 10^8}{3 \times 10^8} \tag{40}$$

$$= 0.7176. \tag{41}$$

$$\boxed{\mathrm{VF} \approx 0.718}. \tag{42}$$

7.4.3 Reflection Coefficient

For the open-ended cable, the measured voltage extrema were

$$V_{\max} = 1.72\,\mathrm{V}, \qquad V_{\min} = 0.060\,\mathrm{V}. \tag{43}$$

The magnitude of the reflection coefficient is given by

$$|\Gamma| = \frac{V_{\max} - V_{\min}}{V_{\max} + V_{\min}}. \tag{44}$$

$$|\Gamma_{\mathrm{open}}| = \frac{1.72 - 0.060}{1.72 + 0.060} \tag{45}$$

$$= \frac{1.660}{1.780} \tag{46}$$

$$= 0.933. \tag{47}$$

$$\boxed{|\Gamma_{\mathrm{open}}| \approx 0.933}. \tag{48}$$

For the $50\,\Omega$ termination, the measured voltage extrema were

$$V_{\max} = 1.09\,\mathrm{V}, \qquad V_{\min} = 0.881\,\mathrm{V}. \tag{49}$$

Thus,

$$|\Gamma_{50\Omega}| = \frac{1.09 - 0.881}{1.09 + 0.881} \tag{50}$$

$$= \frac{0.209}{1.971} \tag{51}$$

$$\approx 0.106. \tag{52}$$

$$\boxed{|\Gamma_{50\Omega}| \approx 0.106}. \tag{53}$$

7.4.4 Characteristic Impedance

For a resistive load, the reflection coefficient is related to the characteristic impedance by

$$Z_0 = Z_L \frac{1 - |\Gamma|}{1 + |\Gamma|}. \tag{54}$$

Using the $50\,\Omega$ termination and the measured reflection coefficient,

$$Z_L = 50\,\Omega, \qquad |\Gamma| = 0.106, \tag{55}$$

we obtain

$$\begin{aligned} Z_0 &= 50\left(\frac{1 - 0.106}{1 + 0.106}\right) && (56)\\ &= 50\left(\frac{0.894}{1.106}\right) && (57)\\ &\approx 40.4\,\Omega. && (58) \end{aligned}$$

Thus, the experimentally estimated characteristic impedance is

$$\boxed{Z_0 \approx 40.4\,\Omega}. \tag{59}$$

7.4.5 Inductance per Unit Length

For a low-loss transmission line,

$$L' = \frac{Z_0}{v_p}. \tag{60}$$

$$Z_0 = 40.4\,\Omega \tag{61}$$

$$v_p = 2.1528 \times 10^8\,\mathrm{m/s}, \tag{62}$$

we obtain

$$\begin{aligned} L' &= \frac{40.4}{2.1528 \times 10^8} && (63)\\ &= 1.876 \times 10^{-7}\,\mathrm{H/m}. && (64) \end{aligned}$$

$$\boxed{L' \approx 1.88 \times 10^{-7}\,\mathrm{H/m}}. \tag{65}$$

7.4.6 Capacitance per Unit Length

Similarly, the capacitance per unit length is given by

$$C' = \frac{1}{Z_0 v_p}. \tag{66}$$

$$C' = \frac{1}{(40.4)(2.1528 \times 10^8)} \tag{67}$$

$$= 1.149 \times 10^{-10}\,\mathrm{F/m}. \tag{68}$$

$$\boxed{C' \approx 1.15 \times 10^{-10}\,\mathrm{F/m}}. \tag{69}$$

8. Results and Discussion

The primary results can be summarized in the table 5

Table 5: Summary of measured and calculated parameters of the coaxial cable.

Quantity	Symbol	Measured/Calculated Value
Cable length	ℓ	0.92 m
Frequency spacing	Δf	117 MHz
Phase velocity	v_p	2.153×10^8 m/s
Velocity factor	v_p/c	0.718
Approximate characteristic impedance	Z_0	40.4 Ω
Inductance per unit length	L'	1.88×10^{-7} H/m
Capacitance per unit length	C'	1.15×10^{-10} F/m
Attenuation/length	α/l	1.63–3.82 dB/m

The results that have been obtained in the fig 3 show that periodically at different frequencies we can see consecutive maxima and minima that must corresponding to certain interference conditions.

The experiment demonstrated the propagation and reflection behaviour of a coaxial transmission line and allowed its important electrical parameters to be determined experimentally. The measured phase velocity was 2.153×10^8 m/s, corresponding to a velocity factor of 0.718. Thus, the signal propagates through the coaxial cable at approximately 72% of the speed of light in vacuum. This reduction is consistent with the presence of the dielectric material between the conductors.

A significant difference was observed between the reflection coefficients for the two terminations. For the open-ended cable, $|\Gamma| = 0.933$, indicating strong reflection from the open end. On connecting the 50 Ω termination, the reflection coefficient decreased to $|\Gamma| = 0.106$. This demonstrates the effect of impedance matching, where a termination closer to the characteristic impedance of the cable results in considerably less reflected power.

The characteristic impedance estimated from the measured reflection was approximately 40.4 Ω, which differs from the expected impedance of 50 Ω termination used in the experiment. This difference may arise from experimental uncertainties in measuring the voltage extrema, connector losses, and non-ideal conditions in the cable and measurement setup. Using the measured characteristic impedance and phase velocity, the distributed inductance and capacitance were determined to be 1.88×10^{-7} H/m and 1.15×10^{-10} F/m, respectively.

The measured attenuation also increased with frequency, from 1.63 dB/m at 0.5 GHz to 3.82 dB/m at 2.0 GHz. This indicates that signal losses become more significant as the frequency increases, consistent with the frequency-dependent losses associated with practical transmission lines.

AI Declaration

AI tools (Primarily ChatGPT) were used as an assistive tool during the preparation of this report. The following uses of AI are acknowledged:

1. **Conceptual understanding:** AI was used to clarify the concepts of phase shifts, reflections, and current distribution in quarter-wavelength antennas.
2. **Transmission lines:** AI was used to understand the concept of transmission lines and their distinction from ordinary wires.
3. **Figures:** AI-based image generation was used to create selected schematic and conceptual figures included in the report. Experimental data plots were generated by using python from the measured data.
4. **Verification of calculations:** AI was used to independently verify numerical calculations performed from the experimental observations.
5. **LaTeX formatting:** AI was used to generate LaTeX tables from experimentally obtained data and to assist with their formatting.
6. **Language refinement:** AI was used to improve the clarity, grammar, and professional presentation of the written content.

Declaration by Akshar Kishore